

Ex. YEARS
No.

UNIT 01:

REAL AND COMPLEX NUMBERS

Section "B"

2. If $Z_1 = -4 + 6i$ and $Z_2 = 2\frac{1}{2} - 2i$ then verify that: $\overline{Z_1 - Z_2} = \overline{Z_1} - \overline{Z_2}$	1.6	2025
2. If $Z_1 = 2 + 3i$ and $Z_2 = 4 + 2i$ then verify that $\overline{\left(\frac{Z_1}{Z_2}\right)} = \frac{\overline{Z_1}}{\overline{Z_2}}$	1.6	2024
2. If $Z_1 = -4 + 6i$ and $Z_2 = 2\frac{1}{2} - 2i$ then verify: $\overline{Z_1 - Z_2} = \overline{Z_1} - \overline{Z_2}$	1.6	2023
2. If $z_1 = 2 - 5i$ and $z_2 = 2 + 3i$, then verify that $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$	1.6	2022

Section "C"

UNIT 02:

LOGARITHMS

Section "B"

3. Find the value of $25.753 \times 0.5341 \times 490.8$ by using logarithm.	2.6	2025
3. Find the value of $\frac{99.87}{(18.369) \times (10.785)}$ by using logarithm.	2.6	2024
3. Find the value by using logarithm:	2.6	2023

$(26.62)^{1/2} \times (87.19)^3$			
$\sqrt{69.53}$			
3. Find the value of $\frac{2391 \times 30.72}{23.34}$ by using logarithm.		2.6	2022
Section "C"			

2022

Section "C"

UNIT 03:

ALGEBRAIC EXPRESSION AND FORMULAS

Section "B"

5. Find the value of $125x^3 + y^3$, when $5x + y = 13$ and $xy = 10$

3.2 2025

4. If $x = 3 - 2\sqrt{2}$ then find the value of $x^2 + \frac{1}{x^2}$

3.4 2024

4. Find the value of $216a^3 - 343b^3$, when $6a - 7b = 11$ and $ab = 8$

3.2 2023

5. Find the value of $125x^3 + y^3$, when $5x + y = 13$ and $xy = 10$.

3.2 2022

Section "C"

UNIT 04

UNIT 04:		FACTORIZATION	
Section "B"			
8. For what value of k , -2 is zero of the polynomial $P(x) = x^3 + x^2 - 14x - k$,	4.7	2025	
5. By using factor theorem find the factors of $x^3 + 5x^2 - 4x - 20$	4.8	2024	
5. Find the factor by using factor theorem. $x^3 - 6x^2 + 11x - 6$	4.8	2023	
8. Find the factors by using factor theorem: $x^3 + x^2 - x - 1$	4.8	2022	
Section "C"			
12. Factorize any four of the following:	4.1	2025	
(i) $r^2 + 9rs - 7rs - 63s^2$			
(ii) $x^2y^2 - 0.1xy + 0.0025$	4.1		
(iii) $3x^2 + 4x - 4$	4.2		
(iv) $8x^3 + 36x^2 + 54x + 27$	4.4		

	(v) $a^3b^3 + 27b^6$		
	(vi) $4x^4 + 625$		
022	12. Factorize any four of the following:	4.5	
	(i) $169x^4 - (3t + 4)^2$	4.2	
	(ii) $7x + xz + 7z + z^2$	4.1	2024
	(iii) $8x^3 + 12x^2y + 6xy^2 + y^3$	4.1	
	(iv) $4x^4 + 1$ (Same question with different variable)	4.4	
	(v) $42x^2 - 8x - 2$	4.2	
	(vi) $a^6 - b^6$	4.2	
25	12. Factorize any four of the following:	4.5	
	(ii) $9y^2z + 3xyz - 9xy^2 - 3x^2y$ [Example:2]	4.2	2023
4	{(ii) Factors of the type: $ac + ad + bc + bd$ }	4.1	
3	(iii) $169x^4 - (3y + 4)^2$	4.1	
	(iv) $108x^4 - 256xz^3$ [Example:2]	4.5	
2	(v) $x^2y^2z^2 + 2xyz - 24$	4.2	
	(iv) $64x^3 - 12x^2 + \frac{3x}{4} - \frac{1}{64}$	4.4	
	12. Factorize the following (Any three)		
	(i) $(2x + z)^2 - (2x - z)^2$	4.1	2022
	(ii) $b^4 + b^2 + 1$	4.2	
	(iii) $x^3 - 8y^3$	4.5	
	(iv) $9a^2b + 18ab^2 - 6ac - 12bc$	4.1	
	(v) $10x^2 - 19xy + 6y^2$ [Example:1]	4.2	

UNIT 05:

ALGEBRAIC MANIPULATION

Section "B"

4. For what value of "q" the expression $36x^4 + 36x^3 + 57x^2 + 24x + q$ will be the perfect square? [Example:2]	5.3	2025
6. For what value of p and q the expression $x^4 + 8x^3 + 30x^2 + px + q$ will be the perfect square?	5.3	2024
6. What should be added to $4x^4 + 4x^3 + 17x^2 - 8x + 9$ to make it a perfect square?	5.3	2023

4. For what value of 'a', the $36x^4 + 36x^3 + 57x^2 + 24x + a$, will be the perfect square?
[Example:2]

5.3 2022

Section "C"

UNIT 06: LINEAR EQUATION AND INEQUALITIES

Section "B"

7. Find the solution set of the equation: $\frac{\sqrt{3y+12}}{7} = 3$ 6.1 2025

7. Find the solution set of the following equation:
 $\frac{|2x+3|}{4} + 2 = 7$ 6.2 2024

8. Find the solution set of the following equation:
 $\frac{|2x-3|}{5} - 12 = 5$ 6.2 2023

6. Find the solution set of the equation:
 $\frac{|2x+1|}{3} - 7 = 1$ 6.2 2022

Section "C"

UNIT 07: LINEAR GRAPHS AND THEIR APPLICATIONS

Section "B"

Section "C"

16. Find the solution set of the following simultaneous equations by graphically method. (Find four ordered pairs for each equation)
 $2x + 2 = y$; $y = x - 1$ 7.3 2025

13. Find the solution set of the following simultaneous equations by graphically method. (Find four ordered pairs for each equation)
 $2x = y + 5$; $x = 2y + 1$ 7.3 2024

13. Find the solution set of the following simultaneous equations by graphical method. (Find four ordered pairs for each equation).
 $3x - y = 11$; $x - 9 = 3y$ 7.3 2023

13. Find the solution set of the following equations graphically. (Find three ordered pairs for each equation) $2x = y + 5$; $x = 2y + 1$

7.3 2022

UNIT 08:

QUADRATIC EQUATIONS

Section "B"

6. Find the solution set by quadratic formula:

$$x^2 + x + 1 = 0$$

[Example: 1(ii)]

8.2 2025

8. Solve by using quadratic formula:

$$2x^2 - 5x - 3 = 0$$

8.2 2024

7. Solve by using quadratic formula: $9x^2 = 12x - 49$

8.2 2023

7. Find the solution set by using quadratic formula:

$$10x^2 + 19x - 15 = 0$$

8.2 2022

Section "C"

UNIT 09:

CONGRUENT TRIANGLES

Section "B"

9. If two angles of a triangle are congruent, then the sides opposite to them are also congruent. Prove It

[THEOREM: 9.1.2]

9.2 2022

Section "C"

13. In any correspondence of two triangles, if one side and any two angles of one triangle are congruent to the corresponding side and angles of the other, the two triangles are congruent. Prove it.

[THEOREM: 9.1.1]

9.1 2025

14. In any correspondence of two triangles, if one side of any two angles of one triangle are congruent to the corresponding side and angles of the other, the two triangle are congruent. Prove it.

[THEOREM: 9.1.1]

9.1 2024

14. In a correspondence of two triangles, if three sides of one triangle are congruent to the corresponding three sides of the other then the two triangles are congruent. Prove it.

[THEOREM: 9.1.3]

9.3 2023

14. In any correspondence of two triangles if one side and any two angles of one triangle are congruent to the corresponding sides and angles of the other, the two triangles are congruent. Prove it. [THEOREM: 9.1.1]	9.1	2022
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UNIT 10: PARALLELOGRAMS AND TRIANGLES

Section "B"

10. If two opposite sides of a quadrilateral are congruent and parallel, it is a parallelogram, prove it. [THEOREM: 10.1.2]	10.2	2022
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Section "C"

15. If two opposite sides of quadrilateral are congruent and parallel. It is a Parallelogram. Prove it. [THEOREM: 10.1.2]	10.2	2025
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15. In parallelogram, the opposite sides are congruent, the opposite angles are congruent, the diagonals bisect each other. Prove it. [THEOREM: 10.1.1]	10.1	2024
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15. If two opposite sides of a quadrilateral are congruent and parallel, it is a parallelogram. Prove it. [THEOREM: 10.1.2]	10.2	2023
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UNIT 11: LINE BISECTORS AND ANGLES BISECTORS

Section "B"

10. The right bisectors of the sides of a triangle are concurrent. Prove it. [THEOREM: 11.1.3]	11.1	2025
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11. Any point on the bisector of an angle is equidistant from its arms. Prove it. [THEOREM: 11.1.4]	11.3	
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10. The right bisectors of the sides of a triangle are concurrent. Prove it. [THEOREM: 11.1.3]	11.1	2024
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11. Any point
from its arms

11.--OR--Any
equidistant

UNIT 12:

9. If two angles
the side opposite
the side opposite

9. The sum of
is greater than

9. Prove that
measure, the
longer than the
Prove that.

UNIT 13:

11. ---OR---
 $m\angle B = 60^\circ$
the bisectors
not required)

11.--OR--Co
 $m\angle P = 50^\circ$
draw the alti

11.--OR--Co
 $m\angle B = 40^\circ$
angle

11. Any point on the bisector of an angle is equidistant from its arms. Prove it

[THEOREM: 11.1.4]

11.--OR--Any point on the bisector of an angle is equidistant from its arms. Prove it.

[THEOREM: 11.1.4]

11.3 2023

11.3 2022

Section "C"

UNIT 12:

SIDES AND ANGLES OF A TRIANGLE

Section "B"

9. If two angles of a triangle are unequal in measure, the side opposite to the greater angle is longer than the side opposite to the smaller angle. Prove it.

[THEOREM: 12.1.2]

12.1 2025

9. The sum of the length of any two sides of a triangle is greater than the length of the third side. Prove it.

[THEOREM: 12.1.3]

12.1 2024

9. Prove that if two angle of triangle are unequal in measure, the side opposite to the greater angle is longer than the side opposite to the smaller angle. Prove that.

[THEOREM: 12.1.2]

12.1 2023

Section "C"

UNIT 13:

PRACTICAL GEOMETRY-TRIANGLE

Section "B"

11. ---OR ---Construct a triangle ABC in which $m\overline{BC} = 6cm$, $m\overline{AC} = 4cm$ and $m\overline{AB} = 5cm$. Draw the bisectors of $\angle A$ and $\angle B$ (steps of construction are not required).

13.2 2025

11.-- OR-- Construct a triangle PQR in which $m\overline{PQ} = 5.7cm$, $m\overline{QR} = 6.4cm$ and $m\overline{PR} = 4.4cm$, draw the altitude from vertex Q and R .

13.2 2024

11.--OR---Construct a $\triangle ABC$ in which $m\overline{BC} = 6cm$, $m\overline{BC} = 4cm$ and $m\overline{AB} = 5cm$, draw the bisectors of angle A and B .

13.2 2023

11. Construct a ΔPQR , in which $m\angle P = m\angle Q = 4.7^\circ$ and $m\angle R = 55^\circ$ [Example:3, Case III]

13.1 2022

Section "C"

UNIT 14:

THEOREMS RELATED WITH AREA

Section "B"

11. Triangles on equal bases and equal altitudes are equal in area. Prove it.

14.1 2024

[THEOREM: 14.1.4]

10. Triangles on equal bases and equal altitudes are equal in area. Prove it.

14.1 2023

[THEOREM: 14.1.4]

Section "C"

UNIT 15:

PROJECTION OF A SIDE OF A TRIANGLE

Section "B"

Section "C"

14. In any triangle, the square on the side opposite to an acute angle is equal to the sum of the squares on the sides containing that acute angle diminished by twice the rectangle contained by one of those sides and the projection on it of the other. Prove it.

15.2 2025

[THEOREM: 15.1.2]

16--OR--In any triangle, the sum of square on any two sides is equal to twice the square on half of the third side together with twice the square on the median which bisect the third side, (Apollonius theorem)

15.2 2024

[THEOREM: 15.1.3]

16. --OR--In an obtuse-angled triangle, the square on the side opposite to the obtuse angle is equal to the sum of the squares on the sides containing the obtuse angle together with twice the rectangle contained by one of the sides, and the projection on it of the other.

15.1 2023

[THEOREM: 15.1.1]

14. --OR-- Prove that the points $P(-3, -4)$, $Q(2, 6)$, and $R(5, 2)$ are collinear.

16. A circle has a diameter AB of length 10 units. A point P on the circumference of the circle is such that $AP = 6$ units. Find the radius of the circle.

16. Show that the points $P(-3, -4)$, $Q(2, 6)$, and $R(5, 2)$ are collinear.

14. --OR-- Show that the points $P(-3, -4)$, $Q(2, 6)$, and $R(5, 2)$ are the vertices of a right-angled triangle.

UNIT 16:

INTRODUCTION TO COORDINATE
GEOMETRY/ ANALYTICAL GEOMETRY

Section "B"

Section "C"

14. ---OR --- Prove that the points $P(-3, -4)$, $Q(2, 6)$ and $R(0, 2)$ are Collinear.	16.2	2025
16. A circle has a diameter between the points $A(-3, 4)$ and $B(11, 6)$. Find the mid-point (centre) and radius of the circle.	16.3	2024
16. Show that the points $P(-3, -4)$, $Q(2, 6)$ and $R(0, 2)$ are collinear.	16.2	2023
14. ---OR--- Show that $(2, 1)$, $B(5, 1)$ and $C(2, 6)$ are the vertices of a right angled triangle. (Example:3, Topic 16.2.3)	16.2	2022

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